# BHARGAVA MANDAPATI


**Senior Software Engineer | Android Application Stack, Framework & Core Architecture**

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## PROFESSIONAL SUMMARY


Accomplished **Senior Software Engineer & Technical Lead** with **over 10 years of experience** architecting, developing, and deploying full-lifecycle **Android applications and systems software**. Highly skilled across the **full Android application stack**, combining deep mastery of core app architecture (**Activity, Fragment, Service, BroadcastReceiver, MVVM, Clean Code, Jetpack, Jetpack Compose**) with low-level systems proficiency in **AOSP internals, HAL/VHAL, and JNI**. Proven track record delivering responsive, high-performance UIs, custom dynamic widgets, and complex multi-module applications integrated with robust backend protocols via **TCP/IP and Protocol Buffers**. Adept at driving end-to-end software quality through rigorous **Espresso UI automation, instrumentation testing**, and system-level performance tuning across CPU, GPU, and input responsiveness.


## CORE COMPETENCIES & TECHNICAL SKILLS

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| **Languages:** | **Java, Kotlin, C / C++ (Systems/Modern C++ concepts)**, XML |
| **Android App Development:** | **Full Android Application Stack** (Activity, Fragment, Service, BroadcastReceiver), **Jetpack Compose, Android Jetpack, MVVM Architecture, Clean Code**, Custom Views & Widgets, Dynamic UI Generation, Runtime Resource Overlays (RRO), UI Theming & Styling |
| **Framework & AOSP:** | **AOSP Internals, Android Framework Services, JNI**, HAL / Custom VHAL, System Bring-Up, SEPolicy & Security Contexts, Inter-Process Communication (IPC/AIDL) |
| **Networking & Middleware:** | **TCP/IP Socket Communication, Protocol Buffers (Protobuf)**, Vehicle Signal Specification (VSS), Franca IDL, SOME/IP, Client-Server & FSA Architecture |
| **Performance & Debugging:** | Performance Benchmarking, UI/Video Latency Tuning, **Dumpsys, Traceview, Android Profiler (CPU / GPU / Memory / Input / Network)**, Logcat Analysis, Bug Triaging |
| **Testing & Build Systems:** | **Gradle, Android App Bundles (AAB), Android Test Infrastructure** (Espresso UI Automation, Instrumentation Testing, Unit Testing), CI/CD fundamentals |
| **Processes & Tooling:** | **Git, Perforce, Jira, Confluence**, Agile / Scrum Methodologies, **SAFe® 5 Practitioner Certified**, Technical Architecture & Code Reviews |

## PROFESSIONAL EXPERIENCE

### System Developer Associate Manager (Team Lead)
**04/2026 – Present**

**Accenture | Bengaluru, India**

- Lead cross-functional engineering teams in the architecture, implementation, and maintenance of modern Android mobile and automotive software applications, establishing high code quality and architecture standards.
- Direct end-to-end feature delivery, proactively mitigating blockers, coordinating cross-team dependencies, and ensuring predictable milestone releases.
- Spearhead project architecture modernization leveraging Kotlin, Java, and modern reactive patterns to guarantee high-performance UI rendering and robust system reliability.

### Team Lead
**11/2022 – 03/2026**

**People Tech Group Inc | Hyderabad, India**

- **Engineering Leadership:** Led a 15-member cross-functional engineering team delivering full-stack Android applications, middleware, and automotive infotainment features for production platforms (General Motors / Cadillac).
- **Real-Time Networking & Protocols:** Architected and implemented low-latency cluster-to-center-stack synchronization transmitting rich UI/media state via **TCP/IP and Protocol Buffers (Protobuf)**.
- **AOSP & Framework Architecture:** Guided engineers in AOSP bring-up, custom HAL development, and framework layer integration; architected custom VHALs and system services for high-throughput data propagation.
- **Multi-Variant Scalability:** Governed multi-variant app and OS feature rollouts, balancing portability, rendering performance, and memory footprint across varied hardware configurations.
- **Developer Tooling & Acceleration:** Designed and delivered custom simulation tools (C++ / Qt) and data format converters to accelerate validation and reduce engineering turnaround time.
- **Client Leadership Delivery:** Traveled to Shanghai to collaborate directly with OEM client stakeholders, resolving critical architectural roadblocks and securing launch success.

### Senior Software Engineer
**08/2021 – 11/2022**

**People Tech Group Inc | Hyderabad, India**

- **C/C++ to Android Application Migration:** Spearheaded migration of legacy RTOS C/C++ control systems into native Android application components, enabling direct touch/UI interactions and hardware abstraction.
- **Dynamic UI Framework & Architecture:** Designed and implemented dynamic UI architectures generating interactive cards and widgets on-the-fly based on real-time signal streams, utilizing MVVM patterns and Runtime Resource Overlays (RRO).
- **Full-Stack Application Controls:** Built virtual control features (Drive & Park, Doors, Windows, Lights) using CarPropertyManager, CAN Manager, and vendor properties integrated with Android services.
- **Testing Automation & Quality:** Authored comprehensive **Espresso UI automation and instrumentation suites**; engineered a dedicated Validator app to ensure middleware communication and UI stability.

### Software Engineer
**07/2018 – 07/2021**

**People Tech Group Inc | Hyderabad, India**

- **AOSP Bring-Up & Custom Hardware:** Contributed to native Android OS bring-up on proprietary automotive boards (CSM), developing custom HALs and authoring SEPolicy security contexts.
- **Multi-Display & Peripheral Integration:** Integrated cluster displays, Head-Up Display (HUD), camera, and rear-view camera (RVC) modules into the central Android infotainment stack.
- **System Profiling & Stability:** Diagnosed application crashes, memory leaks, and input/GPU latency bottlenecks using Android debugging tools and Dumpsys.
- **Code Reviews & Agile Delivery:** Actively participated in peer design reviews, code reviews, sprint planning, and cross-site technical syncs with onsite engineering teams.

### Junior Software Engineer
**04/2016 – 06/2018**

**People Tech Group Inc | Hyderabad, India**

- **Core Android App Development:** Built production-grade Android infotainment applications (Home, Audio, Climate, Navigation, Trailer, Favorites, Phone, Settings) across the full Android application stack.
- **Custom UI Libraries & Dynamic Rendering:** Co-developed the Multi-Functional Control (MFC) library for hardware controller widget navigation, and created custom Skew and Blur graphics libraries for custom freeform displays.
- Delivered robust prototypes that validated Android adoption for General Motors production programs (Info 3.x / 4.0).

## KEY PROJECTS

**SDV Android (Software-Defined Vehicle Framework)** | 03/2025 – 03/2026

- Engineered custom HALs, system services, and Car Services bridging Vehicle Signal Specification (VSS) standards with the Android middleware stack.
- Built a cross-platform C++/Qt simulation tool with dynamic UI generation to accelerate signal validation; integrated SOME/IP, Franca IDL, and Eclipse Kuksa Data Broker on hypervisor-based AAOS environments.

**ClusterService RHMI – High-Speed TCP/IP & Protobuf Pipeline** | 07/2024 – 04/2025

- Delivered an Android system service transmitting real-time UI/data packets (Audio, Phone, Navigation cards) from Center Stack to Instrument Cluster using Protocol Buffers over TCP/IP sockets under Fast Software Architecture (FSA).
- Led a 10-member team of developers and QA, maintaining seamless performance across multi-variant screen resolutions and configurations.

**Project Guava – AOSP Bring-Up on Proprietary Hardware** | 11/2019 – 02/2021

- Customized end-to-end AOSP source tree (HALs, SEPolicy permissions, framework layers) to enable full platform functionality on proprietary automotive computing boards.

## EDUCATION & CERTIFICATIONS

**Education:** **Bachelor of Technology (B.Tech.)** in Electronics & Communications Engineering
Jawaharlal Nehru Technological University, Kakinada (08/2010 – 04/2014)

**Certifications:**
- **Certified SAFe® 5 Practitioner** – Scaled Agile, Inc.
- **The Complete Guide to Becoming a Software Architect** – Udemy
- **AUTOSAR Architecture (Learn from Scratch with Demo)** – Udemy

**Honors & Awards:** Service Anniversary Award | Unmatched Dedication Award | Outside the Box Thinker